

Pedaver / PQNK — Knowledge Outreach & AI Discoverability Audit

Ask PQNK's automated backend is paused. This audit evaluates whether pedaver.com — as it stands today, live in production — gives conventional search engines and AI retrieval systems the greatest legitimate opportunity to find, understand, and cite PQNK's agricultural knowledge. Every finding below is grounded in the live site (fetched directly, not inferred from source), the current codebase, current official crawler/search-engine documentation, and a real 50-query search test.

- Live site verified
- Codebase inspected
- Current docs researched
- 50 real queries tested

1 Executive audit

The technical foundation is genuinely stronger than typical for a site this size. The gap is not crawlability — it's that the site's most important discovery asset, real farmer questions, does not yet exist as published content.

271

URLs in the live
sitemap

186

Knowledge Papers

1 / 50

real farmer queries
where Pedaver
appeared at all

0

notes in every
Farmer Advisory
category, live

ALREADY STRONG

- robots.txt is fully open — no crawler is blocked, no accidental noindex found
- A real, linked JSON-LD entity graph: Organization $\rightleftharpoons$ Person (founder), consistently reused across most page types
- A sophisticated, taxonomy-driven internal-linking engine (fail-fast validated) — a genuine architectural asset
- Clean canonical URLs, good titles/descriptions, consistent Open Graph — verified live
- GA4 and Microsoft Clarity are live and collecting data today
- An extensive, genuinely authored (not thin/scraped) knowledge corpus

WHAT'S ACTUALLY BLOCKING OUTREACH

- The Farmer Advisory page — the site's stated highest-priority discovery asset — is a content-empty placeholder: "0 notes" in all 5 categories, live, today
- The 186-paper corpus was structured around PQNK's own vocabulary, not ordinary farmer phrasing — so even a fully crawlable page rarely matches how a real question is typed
- Google Search Console verification appears broken (code is correct; the deploy secret is likely unset) — this blinds the primary measurement tool
- No deliberate AI-crawler policy exists — training and citation bots are currently allowed identically, by default, not by decision
- Structured-data coverage has real but fixable inconsistencies (detailed in §7-§8)

The central finding: nothing technical is preventing Pedaver.com from being crawled, indexed, or cited. What's missing is the bridge between how PQNK explains its science and how a farmer actually asks about it — and the one page purpose-built to be that bridge currently has no content on it.

2 Current crawl/index status

Verified against the live production site, not inferred from source.

robots.txt — fetched live from pedaver.com/robots.txt

```
User-Agent: * Allow: / Sitemap: https://pedaver.com/sitemap.xml
```

Matches `src/app/robots.ts` exactly — no drift between deployed and current source. Fully open; no crawler-specific rules exist at all (see §3).

sitemap.xml — fetched live

- **271 URLs**, single flat `<urlset>` (not a sitemap index — fine at this scale).
- Generated by `src/app/sitemap.ts`: a hardcoded `staticPages` array plus a map over the `crops/papers/machines/resources` content arrays.
- Every entry has `<lastmod>`, but the semantics are inconsistent: `static/crop/machine/resource` pages get the build's current timestamp (`new Date()` at build time — technically "last build," not "last content change"); paper pages get `paper.publishedDate` directly.
- **Confirmed bug, but a nuanced one**: one entry (`technical-adaptations-for-mechanized-sri-production`) shows `lastmod: 2011-01-09`. This is *not* bad data — the paper's `publishedDate` correctly records its real external publication (Springer, *Paddy and Water Environment* 9(1):111-119, 2011; `externalUrl` and `externalPublisher` fields confirm this is an externally-cited paper, not Pedaver's own authored content). The bug is that `sitemap.ts` uses this field for `lastmod` regardless of whether it's "when Pedaver last touched this page" or "when the cited external work was originally published" — those are different facts, and only the first belongs in a sitemap's `lastmod`.
- No other implausible dates found in this pass (full corpus scan in §7).

Redirects & near-duplicates

`docs/htaccess.txt` defines exactly two 301s: `/farmer-stories/ → /videos/` and `/certification/ → /validation/`. `/videos` and `/video-library` are **not** duplicate content (confirmed: zero overlapping YouTube video IDs between their content sources) — but `/videos` is absent from the main header navigation, reachable only via the sitemap and a single contextual link on crop pages. Not a duplicate-content risk; a discoverability/crawl-depth one.

No hard indexing blocker found — no accidental `noindex`, no broken canonical, no orphaned robots directive.

3 AI crawler accessibility

Sourced from current official documentation (OpenAI, Google, Anthropic, Perplexity, Apple, IndexNow.org), fetched directly — not assumed. See §16 for the exact source list.

CRAWLER	DOCUMENTED PURPOSE	CURRENT PEDAVER.COM TREATMENT
Googlebot	Standard Search indexing	ALLOWED — the single wildcard Allow: / rule covers every crawler below identically. Nothing is blocked; nothing is deliberately chosen either.
Google-Extended	Gemini app training + AI grounding (Google's own docs: <i>"does not impact a site's inclusion in Search nor is it used as a ranking signal"</i>)	
OAI-SearchBot	Powers ChatGPT Search citations	
GPTBot	OpenAI model training	
ChatGPT-User	Live fetch when a person asks ChatGPT to open a specific page (may ignore robots.txt — user-initiated)	
ClaudeBot / Claude-SearchBot	Anthropic training / Claude search-relevance indexing respectively (documented as independently controllable)	
PerplexityBot	Surfaces/links sites in Perplexity search results, not training	
Applebot / Applebot-Extended	Spotlight/Siri/Safari search / Apple Intelligence training respectively	
CCBot	Common Crawl's open archive — widely used as upstream AI-training data by many labs	

OpenAI's own documentation gives the exact configuration for "citable, not trainable" — quoted verbatim from OpenAI's crawler docs: *"a webmaster can allow OAI-SearchBot in order to appear in search results while disallowing GPTBot to indicate that crawled content should not be used for training."* Google, Anthropic, and Apple each publish the equivalent split (Google-Extended, ClaudeBot vs. Claude-SearchBot, Applebot-Extended).

This is a policy decision Pedaver hasn't made yet, not a technical defect. The current config is "allow everyone equally, including training crawlers," by default rather than by choice. §15 (P2) proposes the concrete robots.txt block if a deliberate citation-yes/training-no policy is wanted — not recommended as an automatic default; this is explicitly Pedaver's call.

4 Google readiness

DIMENSION	STATUS	EVIDENCE
Indexability	GOOD	robots.txt open; no noindex meta tag found on live homepage
Sitemap	MINOR ISSUE	Present, fresh mostly, one lastmod-semantics bug (§2)
Canonicalization	GOOD	Verified live: <code><link rel="canonical" href="https://pedaver.com/"></code> , consistent trailing-slash policy
Titles/descriptions	GOOD	Verified live, natural phrasing, not stuffed — e.g. homepage: "PQNK Natural Ecosystem Science of Production Agriculture Pedaver"
Structured data	UNEVEN	Strong core graph, real gaps — full detail §8
Page-level subject clarity	UNEVEN	Strong on Science/Paper pages (rich taxonomy); empty on Farmer Advisory
Original/authored knowledge	GOOD	186 authored papers, not thin or scraped content
Internal relationships	GOOD	Taxonomy-driven, fail-fast-validated linking engine (§10)
Author/organization signals	MOSTLY GOOD	Strong Organization+Person graph; inconsistent on paper pages (§8)
Mobile/performance	NOT INDEPENDENTLY MEASURED	Not covered in this audit's tooling — recommend running Search Console's Core Web Vitals report as a fast next step, not asserted here without data
Duplicate/thin pages	ONE REAL INSTANCE	Farmer Advisory's 5 category pages currently render near-identical "0 notes... check back" placeholder text

Do not build FAQPage schema. Google deprecated FAQ rich results site-wide (deprecation notice May 2025, documentation removed June 2025, effective in Search from May 2026) — confirmed via direct fetch of Google's current Search Central docs. This applies to every site, not just non-authoritative ones. QAPage schema is also not a fit — it's scoped to forum-style, multi-user-answer pages, not Pedaver's single-author knowledge model. Any farmer-question strategy here should be visible, well-structured prose (§11), not schema markup chasing a rich-result feature that no longer exists.

5 Bing/Copilot readiness

DIMENSION	STATUS	NOTE
Bingbot indexing	ALLOWED	Same open robots.txt rule
Sitemap	PRESENT	Same sitemap.xml Bing would consume
IndexNow	NOT IMPLEMENTED	Zero references anywhere in the codebase, confirmed by grep
Bing Webmaster Tools	NO EVIDENCE OF SETUP	No Bing verification signal found
Copilot-specific grounding crawler	NOT DOCUMENTED TO EXIST	Unlike OpenAI/Google/Anthropic/Apple, no official Microsoft doc describes a separate Copilot-grounding bot distinct from Bingbot — flagged as genuinely unverified, not asserted either way

IndexNow, confirmed via the current spec at indexnow.org: a simple ping (HTTP GET or batched JSON POST, up to 10,000 URLs) telling participating engines a URL was added, updated, or deleted — a real-time push instead of waiting for organic re-crawl. Current participants: **Bing, Yandex, Naver, Seznam.cz, Yep** — Google does not participate.

Should IndexNow fire automatically on publish? Yes. The natural hook is the GitHub Actions deploy workflow (`.github/workflows/deploy.yml`) that already runs on every push: a build step could diff the newly-changed content-derived URLs (papers/science/advisory) against the previous deploy and ping IndexNow with just those, at effectively zero ongoing cost or manual effort. Not built in this audit — a concrete, scoped P3 item (§15).

6 ChatGPT Search discoverability

OAI-SearchBot is not blocked — the wide-open robots.txt allows it, so nothing currently prevents pedaver.com pages from being discovered, retrieved, summarized, or cited by ChatGPT Search. This is confirmed against OpenAI's own current crawler documentation, not assumed.

The same wildcard rule also allows **GPTBot** (training) and **ChatGPT-User** (live user-initiated fetch) identically — see §3 for why this is a decision point, not a defect.

Beyond the robots.txt question, ChatGPT Search's actual citation behavior depends on the same fundamentals as conventional search: is the page indexed/crawable (yes), does it contain clear, well-structured, genuinely informative content that answers the query (uneven — strong on Science/Paper pages, absent on Farmer Advisory), and can the retrieval system confidently attribute the content to an authoritative source (helped by the Organization/Person entity graph in §8-§9). No OpenAI documentation was found describing any additional, ChatGPT-Search-specific technical requirement beyond being crawlable and genuinely useful.

7 Knowledge Paper machine-comprehension

Programmatic, corpus-wide inspection of all 186 papers (confirmed count — `src/lib/content/papers.ts` and `src/lib/content/knowledge/metadata.json` agree exactly). Systematic findings, not a per-paper list. One correction to note up front: the discoverability metadata (crops, field problems, questions answered, science domains, authority status, science links) does not live in `papers.ts` itself — it lives in a separate, parallel `metadata.json`, keyed by slug.

The single most important finding in this section: 1,027 real, farmer-phrased questions already exist in the data — and are never shown to a crawler or an AI system. Every one of the 186 papers has a `questionsAnswered` field (3–8 entries each, zero papers with none) — genuine practical questions written in plain language. But `src/app/papers/[slug]/page.tsx` never renders this field on the page, and the paper's JSON-LD never includes it either. The only place these questions are used at all is client-side search ranking (`search.ts`) and the internal relatedness scoring (`related.ts`) — both invisible to anything reading the page's actual content, whether a search crawler or an AI retrieval system. This is a direct, evidenced explanation for several of the 50-query benchmark misses (§14): the answer to "why does mulch suffocate roots" may already be written as one of these questions, on an already-published, already-indexed page — just never displayed.

DIMENSION	RESULT
Unique titles	100% — 0 duplicates across 186
Summary present, reasonable length	100% PRESENT ; 30 papers (16%) exceed 400 characters (up to 485) — long enough to risk truncation as a meta description
crops tagged	84/186 empty, but cross-checked against category: only 1 is a real gap (motha-grass-dominance-a-pqnk-interpretation, arguably a weed subject, defensible either way) — the rest are philosophy/framework/soil-science papers where being crop-agnostic is correct, not a gap
fieldProblems tagged	183/186 (98%) — 3 empty, all abstract philosophy/economics essays
questionsAnswered present	100% — every paper has 3-8 real questions, 1,027 total — but see the callout above: never rendered, never in structured data
scientificDomains coverage	100% — every paper tagged
scientificDomains taxonomy drift	"ECONOMICS" — 55 OCCURRENCES , the only non-canonical value found anywhere in the corpus (every other taxonomy-governed field — crops, practices, principles, production stages, machinery — is 100% clean). Confined specifically to this one field, not a category-level issue.
publishedDate outliers	1/186 — the 2011 SRI paper (§2), a legitimate external citation, not an error, though worth a small UX label so a reader doesn't mistake it for a stale page
authorityStatus distribution	"Current / Approved" 122 (66%) · "Requires PQNK Review" 54 (29%) · "External Knowledge/Evidence" 10 (5%) · zero non-canonical values
scienceLinks coverage	100% — every paper links to at least one of the 10 Science domains (1-6 each)
Paper-to-paper relationship data	DOES NOT EXIST AS STORED DATA — computed live at render time by related.ts; only 6 of 186 slugs (3.2%) have a persisted editorial relationship (the 3 explicit companion pairs). Everything else is algorithmic and never written to structured data.
category distribution	7 buckets, smallest at 9/186 (5%) — not near-empty

A governance nuance worth carrying into §9: 54 papers (29% of the corpus) are explicitly marked "Requires PQNK Review" — not yet approved current knowledge. A machine reading raw content without respecting this field would treat provisional material as equally authoritative as approved knowledge. This is a legitimate, working governance signal already in the data — it just needs to keep being respected as any new surfacing of this content (e.g. rendering questionsAnswered, per above) gets built.

Two related gaps found in how the data that does exist gets used: the paper detail page's ScholarlyArticle JSON-LD (§8) emits only headline, description, author, publisher, url, image — omitting keywords, about/mentions (which could carry the already-tagged scientificDomains/crops), and citation, despite all of this data already existing in metadata.json. Separately, src/lib/content/crops.ts's crop→paper linking

(crossLinks.ts) matches on paper *titles* containing the crop's name, rather than using the already-structured crops field in metadata.json — a strictly better join key that's already there and already used elsewhere in the codebase, just not here.

8 Structured-data assessment

Verified from source (with file:line citations) and cross-checked live on the homepage, founder, and a Science page.

PAGE TYPE	@TYPE USED	ENTITY LINKAGE
Every page (global)	Organization	Verified live: @id, sameAs → Facebook Pedaver + both YouTube channels, founder → founder @id
Founder	Person	Verified live: correct @id, resolves the Organization's founder reference, worksFor back to Organization
Science domains (×10)	TechArticle	Fully linked via @id to both Organization and founder
Crop detail	Article	Fully linked via @id
Machine detail	TechArticle	Fully linked via @id
Knowledge Paper detail	ScholarlyArticle	Not linked — author/publisher are inline literal objects, not @id references; no datePublished/dateModified despite the data existing
Resource detail	TechArticle (most) / HowTo (one slug)	TechArticle branch linked; the single HowTo branch has no author/publisher at all
About	AboutPage	Faithful, linked to Organization
Homepage	<i>none beyond global Organization</i>	Confirmed live — no WebSite, no SearchAction, no listing of the 10 Science domains as data
Farmer Advisory	<i>none</i>	Confirmed live and in source
Papers / Machines / Crops index pages	<i>none</i>	No CollectionPage/ItemList on any of the three — the pages an AI crawler would use to discover the full inventory of each content type declare nothing structurally

Every claim in the JSON-LD examined is faithful to visible page content — no manufactured or unsupported markup found anywhere.

9 PQNK entity/authority architecture

The chain the site should teach machines: Pedaver → PQNK → Science Domains → Knowledge Papers → Crops → Problems → Practices → Machines → Farmer Advisory → Field Evidence.

LINK IN THE CHAIN	CURRENTLY WIRED?	EVIDENCE
Pedaver ⇌ Founder (Asif Sharif)	YES, CORRECTLY	Organization/Person @id pair resolves live (§8)
Pedaver → Science domains	YES	Each Science page's TechArticle links back to Organization + founder
Science ⇌ Knowledge Papers	YES, WELL	Hand-curated "Science Anchor" per domain plus algorithmic, taxonomy-scored supporting papers — a genuinely sophisticated module (§10)
Knowledge Papers ⇌ each other	YES	Weighted scoring cascade (problem → crop → practice → domain → stage → machinery → principle) plus 3 hand-identified explicit companion pairs
Papers → Crops	ONE-DIRECTIONAL AT BEST	Papers tag crops as metadata; Crop's own data model has no relatedPapers/scienceLinks field to link back
Papers → Machines	ONE-DIRECTIONAL AT BEST	Same gap — Machine's data model has no cross-reference field either
Anything → Farmer Advisory	NOT WIRED — NOTHING TO LINK TO	The page is empty (§11)

The core of the entity graph (Pedaver, Asif Sharif, Science, Papers) is genuinely strong and mutually reinforcing. The chain breaks specifically at Crops/Machines (metadata exists but isn't bidirectionally linked at the data-model level) and at Farmer Advisory (nothing to connect to yet). §15 (P2) proposes closing the Crops/Machines gap; §11 addresses Farmer Advisory directly.

10 Internal-link architecture

Two taxonomy-driven functions in `src/lib/content/knowledge/related.ts` power all cross-linking between Papers and Science pages — no manual link lists beyond 3 hand-identified companion pairs:

- **Paper → related Papers** (used on every paper detail page): weighted overlap scoring — same field problem (highest weight) → same crop → same practice → same Science domain → same production stage → same machinery → shared PQNK principle (tiebreaker) — plus explicit companion pairs that always win.
- **Science → related Papers** (used on all 10 Science pages): one hand-curated "anchor" paper per domain, chosen by a human reviewer with documented reasoning per domain, plus algorithmically-selected supporting papers with a diversity pass (prefers introducing a new crop/problem over repeating one already shown). The anchor selection is fail-fast validated — the build throws if a configured anchor slug is ever removed or de-approved, rather than silently falling back to a wrong pick.

This is a real architectural strength, not boilerplate — it's specifically designed to avoid the "breadth mistaken for relevance" failure mode (a broad multi-topic paper outranking a precisely on-topic one), a lesson visibly baked into the scoring weights and the anchor-curation rationale.

Coverage gap: this engine only operates on Papers and Science pages. Crops, Machines, and (once populated) Farmer Advisory are not part of the same relationship graph — see §9.

Structural vs. visible linking: the relationship data computed by this engine renders as real `<a>` links on the page (good for both crawling and human navigation) but is not *also* expressed as structured data (e.g. no `relatedLink/about` properties in the JSON-LD) — a machine currently has to infer the relationship from link position/anchor text rather than read it as an explicit fact.

11 Farmer-question discoverability

The single most important finding in this audit.

Live, today: every one of the Farmer Advisory page's 5 categories reads "0 notes."

Verified directly from production HTML — Adoption Difficulties, Planting, Water Management, Symptoms & Diagnosis, and General Questions each display: *"This library is being built from real questions and voice notes shared across Pedaver's PQNK WhatsApp groups. Check back as new notes are added."* None of the real farmer questions named in this audit's brief — water not infiltrating, yellowing after rain, weeds through mulch, hardpan, the 4 kg NP transition rate, mulch suffocating roots — currently exist anywhere on the live site in question-and-answer form.

This directly explains a visible pattern in the 50-query benchmark (§14): queries phrased in exactly this kind of ordinary language returned zero Pedaver presence, even on topics — hardpan, mulch mechanics — where PQNK's own technical explanation is genuinely distinctive and already documented in the Science/Paper corpus. The knowledge exists; the question-shaped bridge to it doesn't.

The cheapest, fastest win in this entire audit: a bridge already exists and is already written. §7 found that all 186 Knowledge Papers already carry a `questionsAnswered` field — 1,027 real, plain-language questions in total — that is never rendered on the page and never included in structured data. Before writing a single new sentence, simply *displaying* this already-authored content (as a visible "Questions this paper answers" section, and in the paper's JSON-LD as keywords or within the description) closes a meaningful part of the gap at near-zero editorial cost and zero new-content risk. This should happen first, ahead of authoring anything new.

Recommended approach — content, not markup

Since Google has retired FAQ rich results entirely (§4) and Ask PQNK's automated Q&A system is paused, every lever here is purely editorial/rendering work, fully within this audit's "no automated farmer-answering" constraint:

1. **Render the existing 1,027 `questionsAnswered` entries on their paper pages** (and surface them in JSON-LD) — no new writing required, this is exposure work on content that already exists (§7, §15 P1).
2. **Populate a real, modest first batch of Farmer Advisory entries** using the actual WhatsApp-sourced questions named in this audit's brief — written as genuine content a human would read, in the farmer's own phrasing, answered by Pedaver directly. A first pass of 10-15 authentic entries against the existing category structure would already end the "0 notes" placeholder problem.
3. **Where a genuinely new question isn't already covered by an existing paper's `questionsAnswered` list, embed it as a natural subheading inside the relevant Science page or Knowledge Paper** — e.g. the Water Science page gains a passage genuinely titled "Why is water not going into my soil?" This strengthens pages that are already authoritative rather than only adding new ones, with zero risk of reading as keyword-stuffed, since the questions are real and the answers are the site's own existing science.

All three are P1 items in §15 — deliberately scoped as content and rendering work, not a system to build. Item 1 should come first: it's the same fix in spirit as items 2-3, but the content is already written.

12 YouTube & Facebook integration strategy

The three authenticated Pedaver media sources (Facebook Pedaver, Pedaver PQNK YouTube, Aasif Sharif YouTube) are **already correctly declared** via sameAs in the global Organization JSON-LD, verified live — a real, existing strength that establishes the entity linkage without any further work.

/video-library (47 Urdu farmer lectures, organized by topic) and /videos (official + testimonial content) are both live, distinct, non-duplicate pages — but neither carries VideoObject structured data, and no evidence of transcripts or timestamped subject summaries was found for either.

Recommendation — selective, not exhaustive (per the explicit "don't ingest everything yet" instruction)

- **VideoObject markup** (name, description, thumbnailUrl, uploadDate, embedUrl) for a deliberately small, high-value first set — videos already referenced from crop pages or named in Knowledge Papers, not all 47 lectures at once.
- **Contextual links from the relevant Knowledge Paper to the specific video** (timestamped where the platform supports it), rather than a generic "watch our videos" link — this is what makes the website the canonical hub and YouTube the supporting evidence, matching the stated goal directly.
- **Dedicated website pages only for a handful of flagship videos** (e.g. a major transition case study) where the content genuinely warrants standalone treatment — not a blanket policy of one page per video.
- Full transcription/ingestion of the 47-lecture library is explicitly out of scope for this pass, per instruction — flagged as a P3/future decision, not proposed here.

13 Measurement architecture

TOOL	STATUS	NOTE
GA4	LIVE	Confirmed in live HTML, measurement ID present and firing
Microsoft Clarity	LIVE	Confirmed in live HTML
Google Search Console	LIKELY BROKEN	Code path is complete (layout.tsx reads NEXT_PUBLIC_GSC_VERIFICATION, the deploy workflow passes it through) — no verification meta tag found live, so the GitHub Actions secret is most likely unset. A configuration check, not a code fix.
Bing Webmaster Tools	NO EVIDENCE OF SETUP	Recommend setting up alongside IndexNow (§5)
Bing AI Performance / citation visibility	AVAILABLE ONCE WEBMASTER TOOLS IS SET UP	A feature area within Bing Webmaster Tools, not independently verified in this audit
ChatGPT referral traffic	CAPTURABLE VIA GA4 TODAY	Referrer-domain acquisition reports (chatgpt.com) work with GA4 already live — recommend a saved segment, no new tooling needed
AI citation / grounding-query visibility	NO PLATFORM DASHBOARD EXISTS	No official measurement API was found for this, for any provider — the practical proxy is the periodic rerun of the query benchmark in §14
Indexed-page count, impressions, organic queries	BLOCKED ON GSC FIX	Standard GSC reports, available once verification works
Knowledge Paper / Science-page entrances	AVAILABLE NOW	Standard GA4 page-path reporting
YouTube/Facebook → Pedaver referrals	NEEDS A TAGGING CONVENTION	GA4 will show generic "youtube.com" referral traffic today; per-video/per-post attribution needs UTM parameters added to links placed in video descriptions and Facebook posts

The important metric, per the brief, isn't traffic volume — it's whether PQNK knowledge is increasingly retrieved as an authoritative answer source. No platform currently offers a direct dashboard for that. The 50-query benchmark (§14) is the deliberate, reproducible proxy: rerun it on a fixed cadence and track how many of the 50 queries surface Pedaver, not just how many visits arrive.

14 50-query discoverability baseline

Run live via web search, 2026-08-20, verbatim query text, no paraphrasing needed (no two queries were similar enough to skip). **Caveat, stated up front:** the search tool used is documented as US-region only — a Pakistan-based searcher could see a different picture, especially for Pakistan-qualified queries. This baseline should be read as "what a US-based generic search sees," and re-running it from a Pakistan-based context is recommended as a P3 follow-up, not treated as contradicting this data.

1 / 50

queries where
pedaver.com
appeared at all

44

ordinary-language
queries with zero
Pedaver presence

4

branded/PQNK-term
queries where
Pedaver should
appear and mostly
doesn't

7

Pakistan-specific
queries — Pedaver's
home turf — with
zero presence

Full results

#	QUERY	PEDAVER?	TOP COMPETING SOURCES	WHY (INFERENCE)
1	Why is water not going into my soil?	NO	osera.org, geoponicscorp.com, Wikipedia	Generic gardening sites own this phrasing
2	Why does my crop turn yellow after rain?	NO	bighaat.com, Bayer, Univ. Minnesota Extension	Ag-extension/agribusiness dominance
3	Why are weeds coming through mulch?	NO	gardeningknowhow.com, SF Gate, Grasshopper Gardens	Home-gardening sites own this phrasing
4	How does mulch know the difference between weeds and crop seed?	NO	PMC/NCBI, eOrganic	Academic sources own the mechanistic phrasing
5	Why does PQNK recommend only 4 kg NP during transition?	NO	Irrelevant results — search doesn't recognize "PQNK" here	No indexed page uses this exact term+number combination
6	Why can too much finely chopped mulch suffocate roots?	NO	Penn State, GreenUpside, Rutgers NJAES	US extension content owns "over-mulching"
7	What is hardpan and how do I break it without a tractor?	NO	growerexperts.com, livetoplant.com, Quora	Generic homestead content, despite this being core PQNK subject matter
8	Why is my soil compacted even after plowing?	NO	Cornell, Ohioline (OSU), Purdue Agronomy	Land-grant university extension dominance
9	How much water does wheat actually need per acre?	NO	Montana State Extension, Nebraska CropWatch	Generic global agronomy references
10	Why do my crop roots stay shallow?	NO	Eos.org, ScienceDirect, CropWatch	Technical/research content
11	Can I grow crops without irrigation in Pakistan?	NO	PreventionWeb, Dawn.com, PMC	Journalism/NGO water-scarcity coverage

#	QUERY	PEDAVER?	TOP COMPETING SOURCES	WHY (INFERENCE)
12	Why is my soil losing fertility despite fertilizer?	NO	eos.com, greenlife.co.ke	Generic soil-science explainers
13	How do I increase organic matter in sandy soil?	NO	epicgardening.com, Oregon State Extension	US gardening/extension content
14	What causes stunted growth in cotton after sowing?	NO	Springer, ICL India, AgWeb	Agribusiness/agronomy publishers
15	How do I control pests without chemical pesticides?	NO	Lowe's, UC ANR, Minnesota Dept. of Ag	Consumer home/garden + state ag-dept content
16	Why does continuous tillage destroy soil structure?	NO	Iowa State Extension, Nature, PMC	Academic/extension dominance
17	What is regenerative agriculture — does it work in Pakistan?	NO	ISSI Islamabad, Dawn.com, WWF-Pakistan	Press/NGO coverage of the movement, no farming-solution company (Pedaver included) surfaces
18	How do I transition from conventional to organic farming?	NO	Rodale Institute, USDA/SARE	US organic-certification ecosystem — inherently USDA-flavored query
19	Why does my soil turn hard and cracked in summer?	NO	gardeningmentor.com, hortau.com	Consumer gardening blogs
20	What is the best natural fertilizer for wheat?	NO	Medium, growplant.org, ICL	Generic listicle/blog content
21	How can I reduce water usage in rice farming?	NO	IntechOpen, ResearchGate, MDPI	Global rice-science (SRI/AWD) literature — notable since PQNK builds explicitly on SRI
22	Why is biodiversity important on a farm?	NO	Farming Connect (Wales), Science.org	UK/US sustainability orgs and journals

#	QUERY	PEDAVER?	TOP COMPETING SOURCES	WHY (INFERENCE)
23	What is no-till farming and how do I start?	NO	Farmers.gov (USDA), no-tillfarmer.com	USDA-owned topic in a US search context
24	How do earthworms improve soil health?	NO	Fine Gardening, Bayer, USDA blog	Consumer gardening + agribusiness
25	Why do crops need less fertilizer over time in natural farming?	NO	Carbon Herald, Our World in Data	Sustainability-journalism sites
26	What is soil moisture management?	NO	Vaia, Drought.gov, Iowa State Extension	Definitional/reference sites
27	How does mulching reduce irrigation needs?	NO	Oregon State, Frontiers, Springer	Even a Punjab/India zero-till wheat/mulch study (PMC) outranks Pedaver on this exact overlap topic
28	What is PQNK farming system?	NO (pqnk.org ranks #2)	pqnk.org, maximagrotech.com, pqnktech.com	The brand's own term ranks — on a <i>different domain</i> than pedaver.com; apparent domain fragmentation
29	Paedar Qudratti Nizam Kashatqari	NO	facebook.com/Pedaver (x2), tribune.com.pk	Pedaver's own Facebook ranks; pedaver.com itself does not, for its own full system name
30	Pedaver PQNK	YES — position 7 & 10	Instagram, Facebook, YouTube, maximagrotech.com all rank ahead of it	The only hit in 50 queries — and outranked by Pedaver's own social profiles
31	natural farming Pakistan	NO	infrc.or.jp, Wikipedia, hara.com.pk	Arguably Pedaver's most important category term — absent entirely
32	reduce nitrogen fertilizer wheat Pakistan	NO	CIMMYT, FAO, Nature.com	International ag-research institutions
33	sugarcane organic farming Pakistan	NO	IntechOpen, ISAAA, USDA FAS	Industry/policy and research sources
34	how to fix hardpan without machinery	NO	livetoplant.com, homesteadontherange.com	Same gap as query 7; query also noisy (cosmetics use of "hardpan")

#	QUERY	PEDAVER?	TOP COMPETING SOURCES	WHY (INFERENCE)
35	why does my maize turn yellow	NO	Quora, Pioneer/Corteva, Yara	Global agri-input companies own this across many country TLDs
36	best mulch material for vegetable garden Pakistan	NO	lawnlove.com, hortclub.com.pk	A Pakistani commercial mulch retailer ranks — local sites <i>can</i> rank here, just not Pedaver
37	how to test soil compaction at home	NO	lawnstarter.com, Rutgers NJAES	US lawn-care/extension sources
38	natural pest control for cotton	NO	bioprotectionportal.com, cottongins.org	Industry/science-publisher content
39	why is my rice crop lodging	NO	LSU AgCenter, PMC, riceextension.com.au	Global rice-science/extension sources
40	how much organic matter does soil need	NO	Oregon State, SARE, ResearchGate	University soil-health reference content
41	companion planting for pest control	NO	Wikipedia, IntechOpen	Consumer gardening blogs
42	can jantar sesbania fix soil nitrogen	NO	ztbl.com.pk, IRRI Knowledge Bank, a small Pakistani farm blog	Very PQNK-adjacent, Pakistan-specific, Urdu-vocabulary term — and still no Pedaver
43	how deep should mulch be applied	NO	lawnstarter.com, Virginia Cooperative Extension	US landscaping/extension content
44	why do farmers till their soil	NO	Quora, USDA ERS	General-education and USDA sources
45	climate resilient farming practices Pakistan	NO	minutemirror.com.pk, Frontiers	Pakistani press/NGO/academic coverage — no farming-solutions company surfaces
46	reducing chemical dependency in agriculture	NO	Nature, farmonaut.com	Global sustainability-policy and ag-tech content
47	soil biology and microbial activity explained	NO	biologyinsights.com, ScienceDirect	Science-explainer and extension sources

#	QUERY	PEDAVER?	TOP COMPETING SOURCES	WHY (INFERENCE)
48	what is production architecture in farming	NO	Dezeen, Agritecture, ArchDaily	Naming collision — read as building/vertical-farming architecture, not the PQNK term
49	how to improve water infiltration in clay soil	NO	alluvialsoillab.com, USDA NRCS	Soil-science and ag-input company content
50	field evidence for regenerative farming Pakistan	NO	ISSI Islamabad, Dawn.com/Prism	A Dawn/Prism farmer profile reads almost exactly like a PQNK case study, yet doesn't name Pedaver — a near-miss worth investigating for outreach, not indexing

This exact query list and methodology becomes the baseline — rerun on a fixed cadence, same wording, same caveats noted each time, to measure whether the P0-P3 work in §15 moves the needle.

15 Prioritized opportunity map — P0 through P3

Every item below states the problem, evidence, affected pages, proposed correction, expected benefit, difficulty, and risk. Analysis only — nothing here has been implemented.

P0 — DISCOVERY BLOCKERS

No hard technical blocker (broken robots.txt, accidental noindex, broken canonical) was found anywhere in this audit — worth stating plainly as good news. One item below is P0-adjacent because it blinds measurement of everything else.

P0 Google Search Console verification is likely non-functional

Evidence	Code is correct (layout.tsx:28–30, secret wired through deploy.yml:29); no google-site-verification meta tag found on the live homepage
Affected	Site-wide measurement, not any specific page
Correction	Confirm/set the NEXT_PUBLIC_GSC_VERIFICATION GitHub Actions secret; no source code change needed
Benefit	Unblocks indexed-page counts, impressions, and query data — the entire §13 measurement framework depends on this
Difficulty	Trivial (a repo-settings check, not a code change)
Risk	None

P1 — KNOWLEDGE-UNDERSTANDING IMPROVEMENTS

P1 · DO THIS FIRST

Render the 1,027 already-written farmer questions that currently exist nowhere visible

Problem	Every paper's questionsAnswered field (3–8 real, plain-language questions each) is fully populated but never rendered on the page and never included in JSON-LD
Evidence	Corpus-wide field audit (§7); confirmed via repo-wide search that the field is read only by client-side search/relatedness logic, never by the page template
Affected	All 186 Knowledge Paper detail pages (src/app/papers/[slug]/page.tsx)
Correction	Add a visible "Questions this paper answers" section to the page template; include the same list in the ScholarlyArticle JSON-LD (as keywords or folded into an expanded description)
Benefit	The single highest-leverage fix in this audit — real content, already written, currently invisible to every crawler and AI system; directly closes gaps evidenced in §14
Difficulty	Low — one template change surfaces content for all 186 pages at once, no new writing required
Risk	None — the content already exists and was already editorially approved as part of each paper

P1 Populate Farmer Advisory with real, authentic farmer questions

Problem	The site's stated highest-priority discovery asset is a content-empty placeholder
Evidence	Live page: "0 notes" in all 5 categories, confirmed via direct fetch (§11)
Affected	/advisory
Correction	Hand-author a first batch (10-15) of real WhatsApp-sourced Q&A entries, written naturally, no schema dependency
Benefit	Directly closes the gap responsible for most of the 44/50 zero-presence queries in §14
Difficulty	Moderate — editorial work, not engineering
Risk	Low, provided questions stay authentic (per the explicit "no keyword-stuffing" instruction)

P1

Embed real farmer questions as natural subheadings inside existing Science/Paper pages

Problem	PQNK's technical explanations exist but aren't phrased as the questions farmers actually ask
Evidence	§14 queries 1, 3, 6, 7, 34 — mechanism is documented, ordinary phrasing isn't present anywhere to match against
Affected	Relevant Science pages (Water, Soil) and specific Knowledge Papers on mulch/hardpan
Correction	Add a small number of genuine question-phrased passages to pages that already cover the mechanism — augment, don't create new pages
Benefit	Strengthens already-authoritative pages; helps both conventional and AI retrieval match casual phrasing
Difficulty	Low-moderate — editorial additions to existing content
Risk	Low if genuinely authored, not templated

P1 Fix sitemap.ts's lastmod semantics for externally-cited papers

Problem	publishedDate is used for sitemap lastmod regardless of whether it's Pedaver's own edit date or an external citation's original date
Evidence	The 2011 SRI paper's lastmod (§2)
Affected	src/app/sitemap.ts:45
Correction	Use a separate "page last touched" signal (or the build date, matching other content types) for papers with an externalUrl, reserving publishedDate for its citation meaning
Benefit	Accurate freshness signal to crawlers; currently one externally-cited page understates its own currency
Difficulty	Low — small, contained code change
Risk	None

P1 Fix Knowledge Paper JSON-LD: link author/publisher by @id, add dates

Problem	Every other content type links author/publisher via @id into the shared entity graph; paper pages use inline literal objects instead, and omit datePublished/dateModified entirely
Evidence	src/app/papers/[slug]/page.tsx:71–77 (confirmed via codebase audit)
Affected	All 186 Knowledge Paper detail pages
Correction	Replace inline author/publisher with @id references matching every other page type; surface paper.publishedDate as datePublished
Benefit	Closes the one real inconsistency in an otherwise well-connected entity graph, across the largest single content type on the site
Difficulty	Low — one template change applies to all 186 pages at once
Risk	None

P1

Add CollectionPage/ItemList JSON-LD to the Papers, Machines, and Crops index pages

Problem	The three pages that enumerate the full inventory of each content type declare nothing structurally
Evidence	Confirmed absent in src/app/papers/page.tsx, machines/page.tsx, crops/page.tsx
Affected	3 index pages
Correction	Add CollectionPage/ItemList mirroring the existing detail-page JSON-LD pattern
Benefit	Gives crawlers/AI systems a structured inventory rather than requiring them to infer it from rendered links
Difficulty	Low
Risk	None

P1 Add page-specific JSON-LD to the homepage and Farmer Advisory

Problem	Both rely solely on the sitewide Organization block
Evidence	Confirmed live and in source (§8)
Affected	src/app/page.tsx, src/app/advisory/page.tsx
Correction	Homepage: WebSite (+ optionally an ItemList of the 10 Science domains). Advisory: a faithful WebPage/Service block once real content exists — sequence this after the content work above, not before.
Benefit	Completes structured-data coverage on the two most-linked entry points
Difficulty	Low
Risk	None

P2

Wire Crops and Machines into the same relationship graph as Papers/Science

Problem	The chain Pedaver → Science → Papers → Crops → Machines is only one-directional past Papers — Crop and Machine data models have no cross-reference fields, and where a crop→paper link is computed at all, it uses a weaker method than what's already available
Evidence	<code>src/lib/content/crops.ts/machines.ts</code> interfaces lack <code>relatedPapers/scienceLinks</code> -style fields; separately, <code>crossLinks.ts</code> matches crops to papers by searching for the crop's name inside paper <i>titles</i> , rather than using the already-structured, already-populated <code>crops</code> field in <code>metadata.json</code> — a strictly better join key that's already used elsewhere in the codebase (§7)
Affected	Crop and Machine content models, detail pages, and <code>src/lib/content/crossLinks.ts</code>
Correction	Add explicit cross-reference fields feeding both visible links and JSON-LD, following the pattern already proven in <code>related.ts</code> ; switch <code>crossLinks.ts</code> to key off <code>metadata.json</code> 's <code>crops</code> field instead of title text-matching
Benefit	Completes the full entity chain named in the audit brief; the title-matching fix alone likely catches paper↔crop relationships the current text search misses
Difficulty	Moderate — data-model + template work across two content types
Risk	Low

P2

Resolve the "Economics" taxonomy-drift and any other non-canonical domain values

Problem	A domain/category value used across the corpus (55 occurrences, confirmed) doesn't match any of the 10 canonical Science domains, breaking the Science-page linkage for every paper carrying it
Evidence	Corpus-wide scan of <code>papers.ts</code> (§7)
Affected	Up to 55 Knowledge Papers
Correction	Either fold "Economics" content into the nearest canonical domain(s) or make a deliberate decision to formalize it as an 11th domain with its own Science page — a content decision, not just a data fix
Benefit	Restores taxonomy integrity for a meaningful fraction of the corpus
Difficulty	Moderate — depends on which resolution is chosen
Risk	Low if done as reclassification; a new domain page is a bigger, separate decision

P2 Add /videos to primary navigation

Problem	A live, sitemap-indexed page with distinct content is absent from the header nav, reachable only via sitemap and one contextual link
Evidence	<code>src/components/Header.tsx:73–79</code> lists <code>/advisory</code> , <code>/farmer-voices</code> , <code>/video-library</code> — not <code>/videos</code>
Affected	Header navigation, crawl depth for <code>/videos</code>
Correction	Add a nav entry, or make a deliberate decision that this content should live inside <code>/video-library</code> instead — either resolves the orphan status
Benefit	Improves crawl-path clarity and human discoverability of testimonial/official-video content
Difficulty	Low — but touches the locked header, requires explicit approval per standing instruction
Risk	Low, contained to one nav entry

P2 Make a deliberate AI-training-crawler policy decision

Problem	Training crawlers (GPTBot, ClaudeBot, CCBot, Google-Extended, Applebot-Extended) are currently allowed identically to citation crawlers, by default, not by choice
Evidence	§3, current live <code>robots.txt</code>
Affected	<code>src/app/robots.ts</code>
Correction	If Pedaver wants citation without bulk training exposure, OpenAI/Google/Anthropic/Apple each publish an independent opt-out token for exactly this split (§3) — implementable in a few lines once the policy is decided
Benefit	Turns an accidental default into a deliberate choice
Difficulty	Trivial once decided
Risk	None technically; the only real "risk" is the business decision itself, which is explicitly Pedaver's to make, not a default to assume

P3 – OUTREACH EXPANSION

P3 Implement IndexNow, automated on deploy

Correction	A key-file + ping step added to <code>.github/workflows/deploy.yml</code> , firing only for newly-changed content URLs
Benefit	Faster Bing/Yandex/Naver/Seznam/Yep discovery of new/updated Knowledge Papers and Advisory entries — no effect on Google, which doesn't participate
Difficulty	Low-moderate
Risk	None

P3 Set up Bing Webmaster Tools

Correction Verification, sitemap submission, IndexNow key registration, and review of the AI Performance reporting area once available

Benefit Closes the largest measurement gap after GSC

Difficulty Low

P3 Broader video/Facebook integration, sequenced after the above

Correction Expand VideoObject markup and contextual linking beyond the initial high-value set (§12), once the core content and structured-data work is proven

Benefit Extends the "website as canonical hub" pattern to the full media library

Difficulty Moderate, ongoing

P3 Rerun the 50-query benchmark, including from a Pakistan-based context

Correction Same query list and methodology (§14), plus a second pass from a Pakistan-located tool/VPN to address the US-only search-scope caveat

Benefit The actual, ongoing measurement of whether this work is succeeding

Difficulty Low, repeatable

16 Exact files that would eventually need modification

FILE	CHANGE
src/app/sitemap.ts	Fix lastmod semantics for externally-cited papers (P1)
src/app/papers/[slug]/page.tsx	Render the existing questionsAnswered field visibly; link author/publisher by @id; add datePublished/dateModified/keywords/about to JSON-LD (P1 — highest leverage item in this audit)
src/lib/content/crossLinks.ts	Switch crop→paper matching from title text-search to the structured crops field in metadata.json (P2)
src/app/resources/[slug]/page.tsx	Add author/publisher @id to the HowTo branch (P1)
src/app/papers/page.tsx	Add CollectionPage/ItemList JSON-LD (P1)
src/app/machines/page.tsx	Add CollectionPage/ItemList JSON-LD (P1)
src/app/crops/page.tsx	Add CollectionPage/ItemList JSON-LD (P1)
src/app/page.tsx	Add WebSite (+ optional ItemList of Science domains) JSON-LD (P1)
src/app/advisory/page.tsx	Populate real content; add JSON-LD once populated (P1)
src/lib/content/crops.ts	Add relatedPapers/scienceLinks-style cross-reference fields (P2)
src/lib/content/machines.ts	Same (P2)
src/lib/content/papers.ts	Reclassify the ~55 "Economics"-tagged entries once a resolution is chosen (P2)
src/components/Header.tsx	Add /videos nav entry, or fold its content into /video-library — requires explicit approval (locked nav) (P2)
src/app/robots.ts	Add deliberate per-crawler rules, once the training-vs-citation policy is decided (P2)
.github/workflows/deploy.yml	Add an IndexNow ping step for changed content URLs (P3)
GitHub repo secrets	Verify/set NEXT_PUBLIC_GSC_VERIFICATION — not a source file (P0)
Relevant Science pages / Knowledge Papers (specific set TBD)	Add real farmer-question passages (P1) — exact list to be chosen editorially, not templated

**Audit and report only. No file above has been touched.
Awaiting approval before any implementation work begins.**

